

BEAUFORT COUNTY AP, SC, USA

WMO: 720612

Lat: 32.412N

Lon: 80.634W

Elev: 38

StdP: 100.87

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00203

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	0.3	-10.9	1.5	2.4	-7.7	2.0	4.4	10.4	8.1	9.3	9.6	3.6	10	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	34.1	25.1	32.8	24.9	32.2	24.9	26.9	30.8	26.5	30.2	26.0	29.7	3.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	28.4	25.2	20.5	27.9	25.1	20.3	27.9	84.6	31.5	82.9	29.9	80.9	29.6	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.0	7.1	-2.7	36.3	2.0	0.8	-4.1	36.9	-5.3	37.4	-6.4	37.8	-7.9	38.4		
			-4.9	27.7	1.7	0.6	-6.2	28.2	-7.2	28.5	-8.1	28.9	-9.4	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	9.9	12.1	14.6	19.4	23.5	27.1	28.1	27.6	25.5	20.7	14.8	12.3
	DBStd	7.33	4.98	4.73	4.42	3.35	2.69	2.07	1.64	1.54	2.12	3.60	4.14	4.57
	HDD10.0	142	64	29	12	0	0	0	0	0	0	7	29	29
	HDD18.3	933	263	177	129	28	2	0	0	0	19	122	192	192
	CDD10.0	3672	61	89	155	281	418	512	561	546	466	333	150	101
	CDD18.3	1420	2	4	13	59	162	262	302	287	216	93	14	6
	CDH23.3	12554	3	16	65	283	1062	2617	3298	2933	1711	526	34	5
	CDH26.7	4206	0	0	9	33	258	978	1302	1046	493	87	1	0
Wind	WSAvg	2.9	3.1	3.1	3.3	3.2	2.9	2.5	2.5	2.3	2.6	2.8	3.0	3.0
Precipitation	PrecAvg	1209	84	82	90	74	71	129	140	162	121	116	64	75
	PrecMax	1720	220	283	218	182	183	320	256	413	235	532	142	266
	PrecMin	811	15	8	8	14	12	38	30	51	17	4	0	26
	PrecStd	216	55	56	48	47	49	64	60	76	54	101	45	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	25.2	27.8	30.0	32.7	36.0	35.9	35.1	32.6	30.2	26.2	23.9
		MCWB	18.4	18.4	17.6	19.7	21.6	25.0	25.4	25.8	24.3	24.0	20.8	20.4
	2%	DB	21.2	22.8	25.1	27.4	31.1	34.0	34.0	32.9	31.3	28.7	24.1	22.4
		MCWB	17.3	17.5	17.8	20.2	21.7	24.6	25.5	25.5	24.3	23.7	19.9	19.5
	5%	DB	18.9	21.3	22.7	26.2	29.1	32.6	32.7	32.3	30.9	27.5	22.6	21.1
		MCWB	15.5	16.7	16.7	19.5	22.0	24.5	25.4	25.4	24.1	22.2	19.0	19.2
	10%	DB	17.4	19.1	21.3	25.1	27.7	31.3	32.1	31.2	29.9	26.3	21.3	19.8
		MCWB	14.8	15.8	16.3	19.0	21.6	24.3	25.3	25.1	23.8	21.4	18.5	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	20.5	20.8	23.2	24.7	27.4	27.5	27.4	26.7	25.7	23.2	22.3
		MCDB	21.6	23.0	23.7	26.1	28.6	31.8	31.8	30.7	29.8	27.7	24.8	23.4
	2%	WB	18.0	19.1	19.4	22.3	23.8	26.5	26.9	26.9	25.9	24.6	21.7	21.1
		MCDB	20.0	21.9	22.8	25.2	27.6	30.9	31.1	30.3	28.6	27.4	22.6	21.7
	5%	WB	17.0	17.8	18.3	21.4	23.2	25.8	26.4	26.4	25.5	23.7	20.4	19.5
		MCDB	18.1	19.7	21.4	24.2	26.6	30.1	30.3	30.0	28.3	26.8	21.8	20.4
	10%	WB	15.6	16.6	17.4	20.6	22.7	25.3	25.9	25.9	24.9	22.8	18.9	17.6
		MCDB	17.2	18.6	19.6	23.3	26.1	29.6	29.8	29.3	27.9	25.4	21.3	18.6
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.9	9.8	9.2	8.3	8.1	7.8	7.4	7.5	8.8	9.4	8.9
		MCDBR	10.7	11.0	11.2	9.8	9.8	9.5	8.6	8.5	8.4	8.8	10.0	9.5
		MCWBR	7.1	6.9	5.9	4.4	3.6	3.8	3.6	3.4	3.0	4.3	6.0	6.5
	5% WB	MCDBR	9.0	9.6	9.5	7.8	7.7	8.3	7.9	7.8	7.2	7.6	7.6	8.1
		MCWBR	7.6	7.1	5.9	4.3	3.3	3.9	3.7	3.5	3.1	4.3	5.8	6.8
Clear-Sky Solar Irradiance	taub	0.330	0.339	0.369	0.402	0.454	0.493	0.522	0.505	0.429	0.394	0.353	0.338	0.338
	taud	2.534	2.505	2.441	2.340	2.234	2.173	2.107	2.156	2.364	2.431	2.502	2.539	2.539
	Ebn at Noon	893	918	911	889	842	805	779	788	839	845	856	862	862
	Edh at Noon	84	95	109	126	141	150	159	150	117	102	87	80	80
All-Sky Solar Radiation	RadAvg	2.93	3.59	4.70	5.97	6.39	6.41	6.24	5.65	4.79	4.18	3.26	2.60	2.60
	RadStd	0.16	0.36	0.43	0.38	0.42	0.46	0.38	0.37	0.40	0.44	0.22	0.21	0.21

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.60	+0.76	N/A	N/A	N/A	N/A	N/A	-98	+208	+144	

Nomenclature: See separate page